



ME 342A: SYSTEM DESIGN AND ANALYSIS

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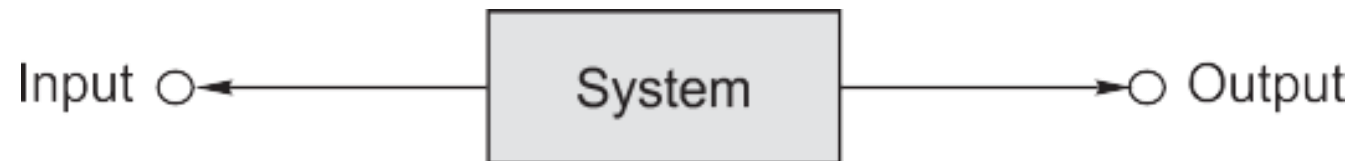
Unit 1 System Fundamental Concepts

Classification of Systems with examples



1. Open-loop and closed-loop systems

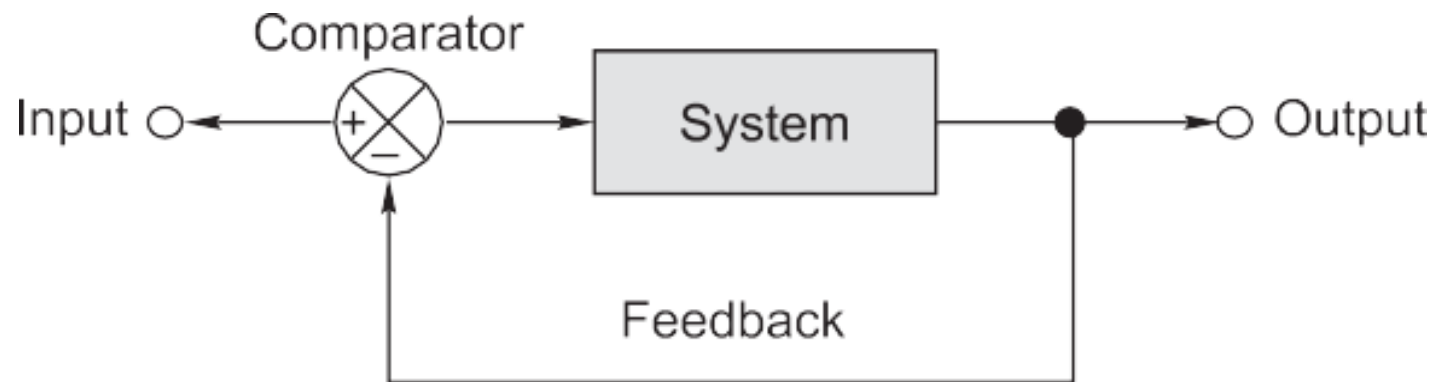
The systems in which output has no effect on the control action are called **open-loop systems**.



(a) Open-loop system



Feedback control systems are referred to as **closed-loop systems**.



(b) Closed-loop system



2 Linear and non-linear systems

The output should be zero for zero input.

There should not be any non-linear operator present in the system.

Examples of non-linear operators:

Trigonometric operators: Sin, Cos, Tan, Cot, Sec, Cosec etc.



Exponential, logarithmic, square, Cube etc.

Either input x or output y should not have these non-linear operators.

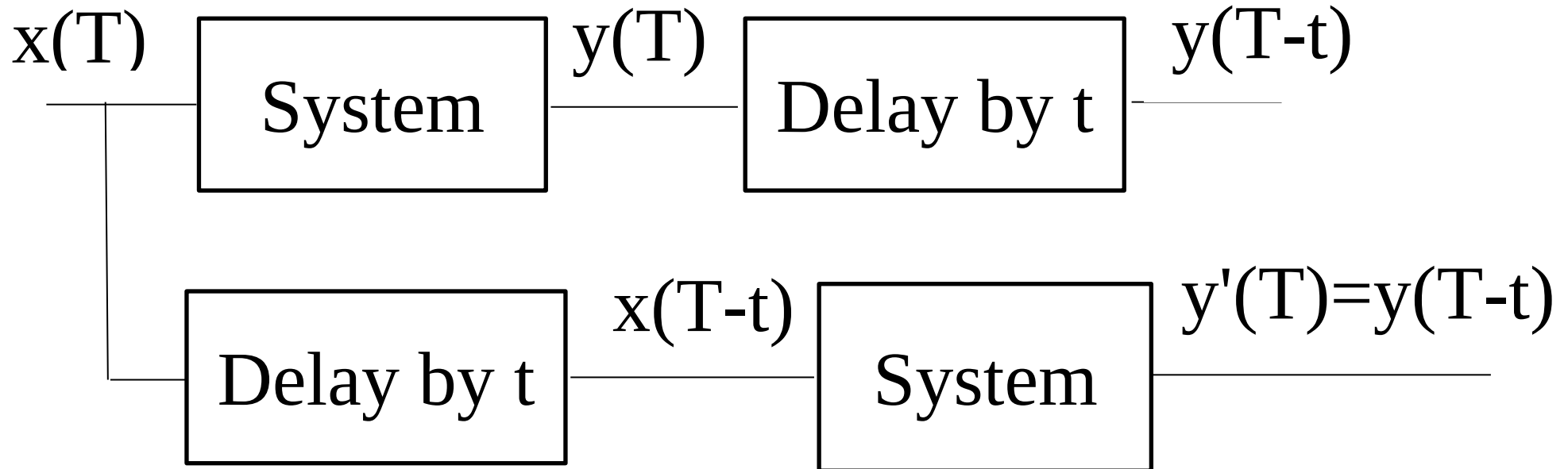
Examples: Linear System: $y(t) = \sin x(t)$

Examples: Non-linear System

$$y(t) = x(t) + 3; \quad y(t) = \sin(x(t))$$



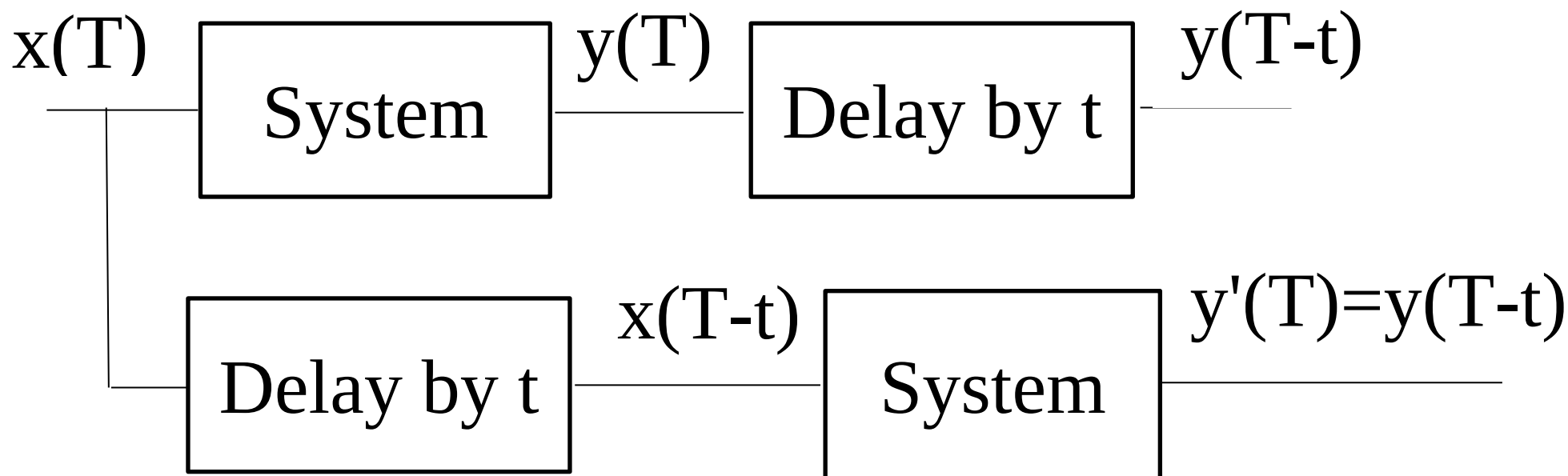
3 Time-invariant and time-varying systems



Time varying: $y(T) = x(2T)$

a. $x(2T-2t)$

b. $x(2T-t)$



Time invariant: $y(T) = \sin[x(T)]$

a. $\sin x(T-t)$ b. $\sin x(T-t)$

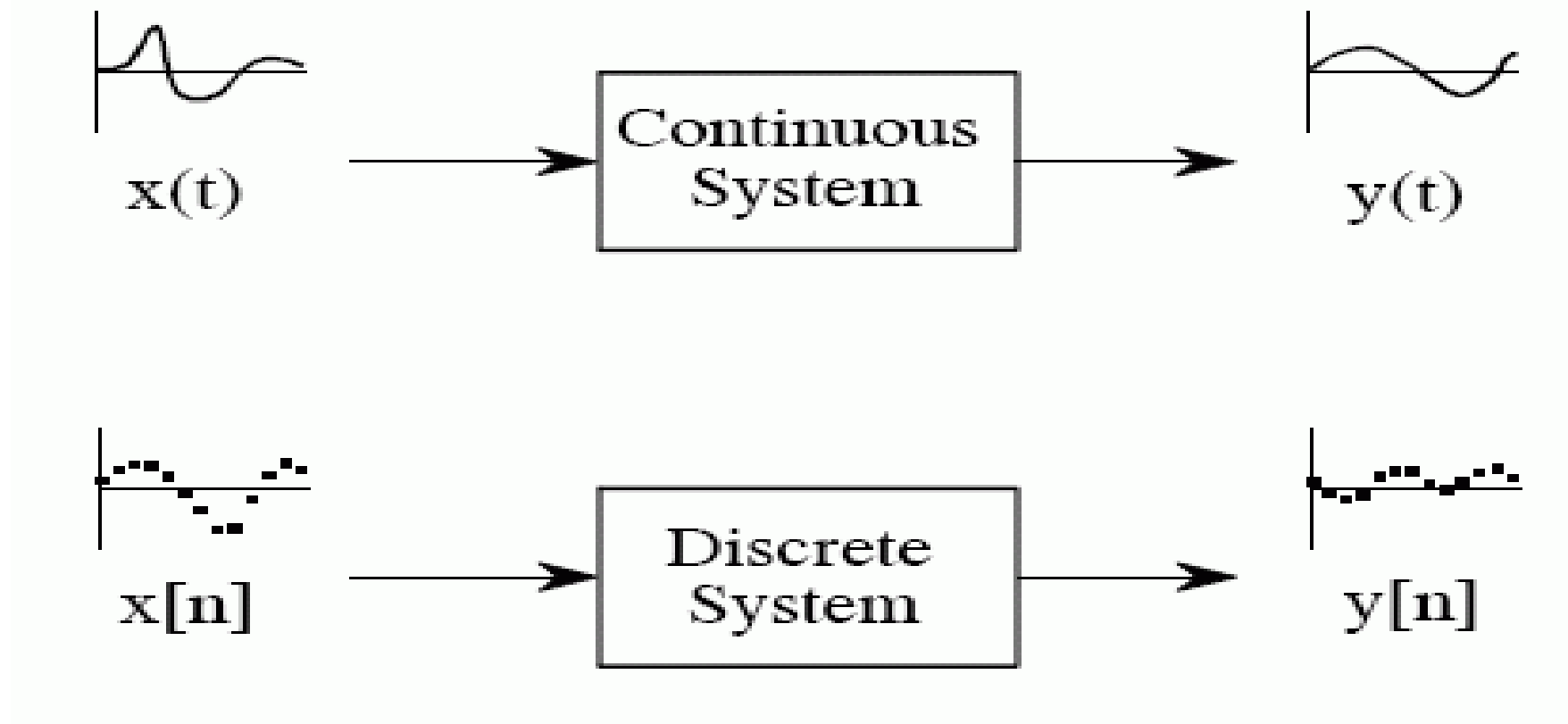


4 Continuous time and discrete systems

Speed control of a DC motor: Output, rotation in terms of radians per second is a function of voltage and current provided as input to it on a continuous time basis. Such a system takes a continuous time input and provides a continuous time output.

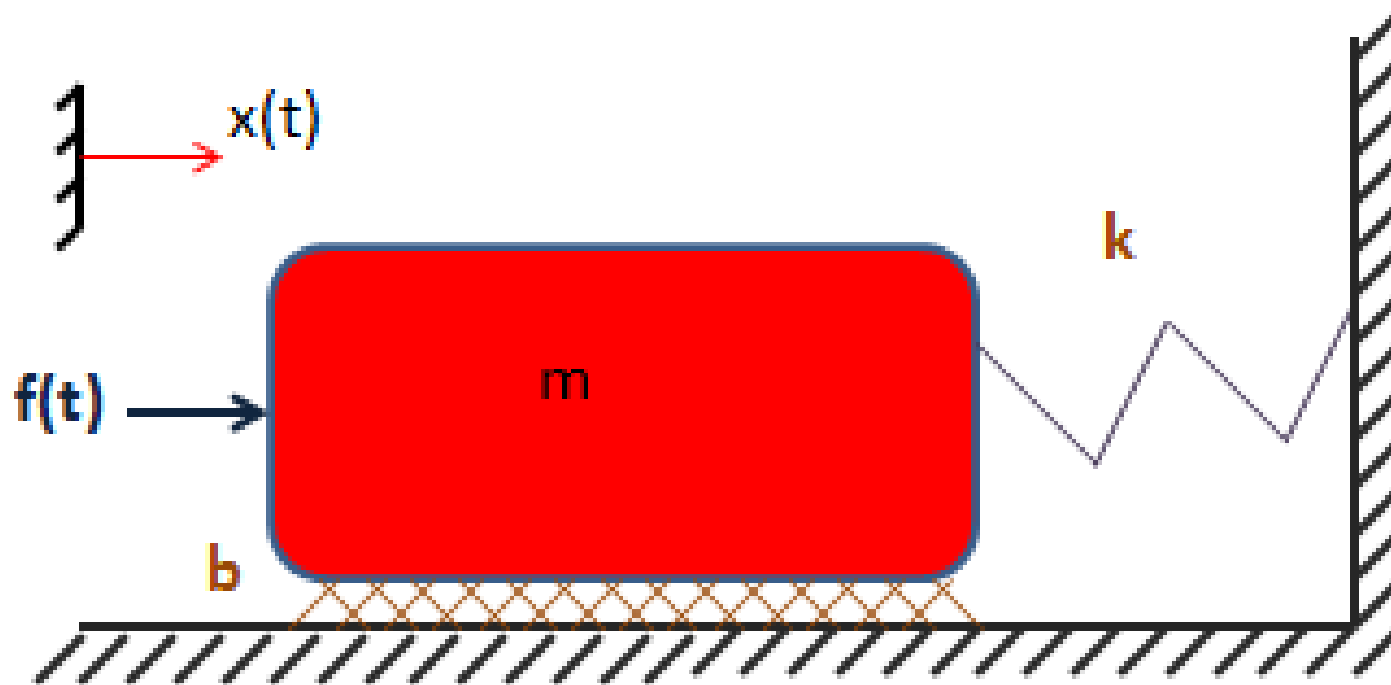


Generator excitation system: Field current is adjusted based on feedback from the output voltage generated to keep the output voltage constant.



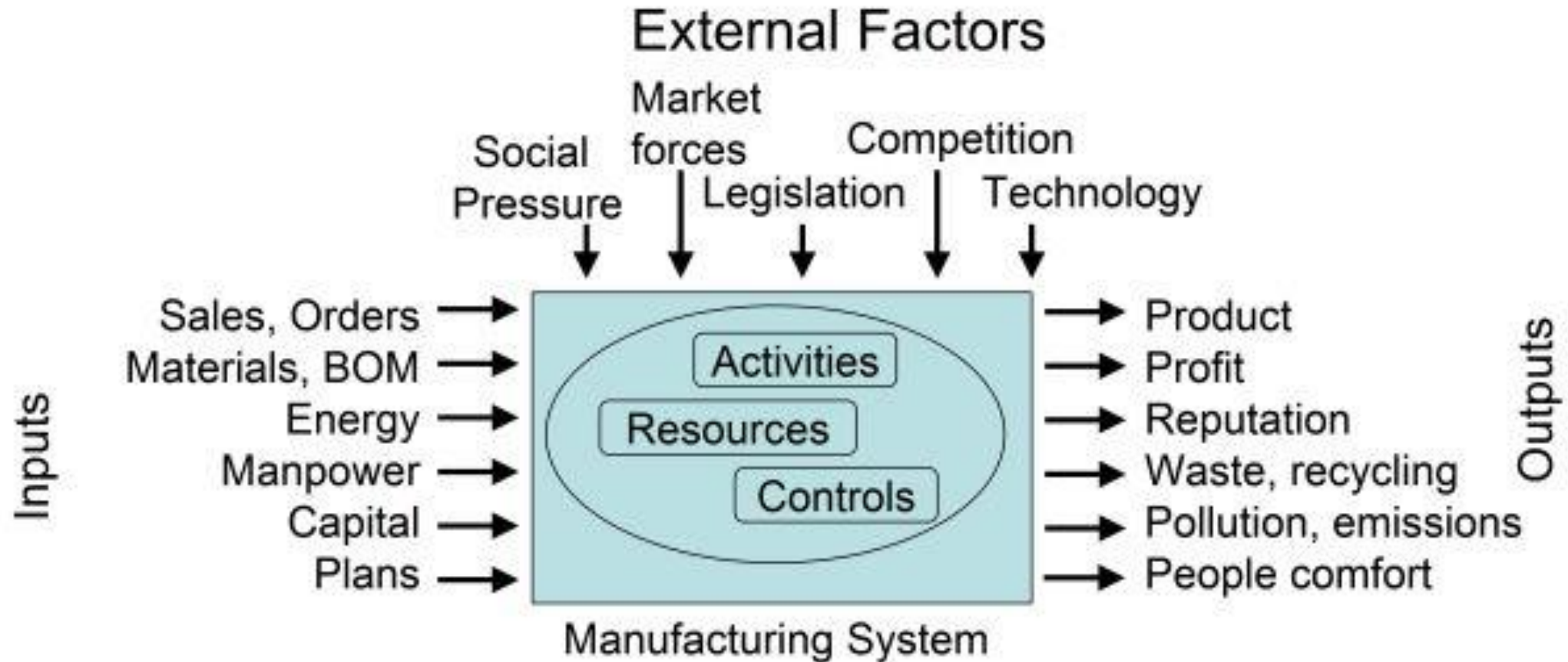


5 Single-Input–Single-Output (SISO) systems





5 Multi-Input–Multi-Output (MIMO) systems





6 Lumped parameter and distributed parameter systems

The elements in lumped system are **concentrated at singular points** in space. Example: Electrical circuit with resistor, inductance, and capacitor. Current and voltage are functions of **time (only)**.



The elements in distributed systems are **distributed in space**. Example: Electrical line where inductance, capacity and resistance are not constant but functions of length. Current and voltage **depend on both time and space**.



7 Deterministic and stochastic systems

Deterministic Simulation Model does not contain any probabilistic components

- Example: a system of differential equations representing a chemical reaction
- Output are also deterministic

Stochastic Simulation models are those having at least some random input components

Examples include Queuing models (Interarrival times between two consecutive customers and service times are usually random)

They produce output that are also random



8 Static and dynamic systems

Static Simulation Model is a representation of a system at a particular point in time (i.e., time plays no role)

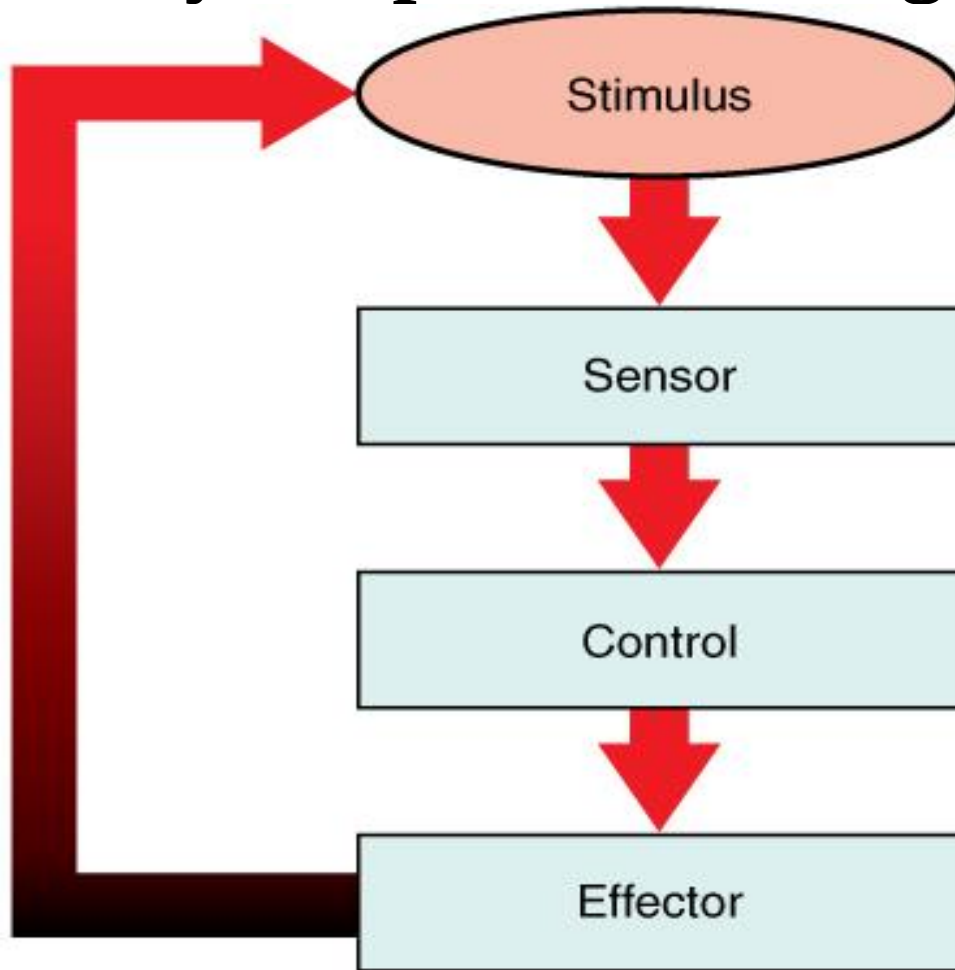
- o Estimating the probability of winning a soccer game
- o Estimating the value of π

Dynamic Simulation Model is a representation of a system as it evolves over time

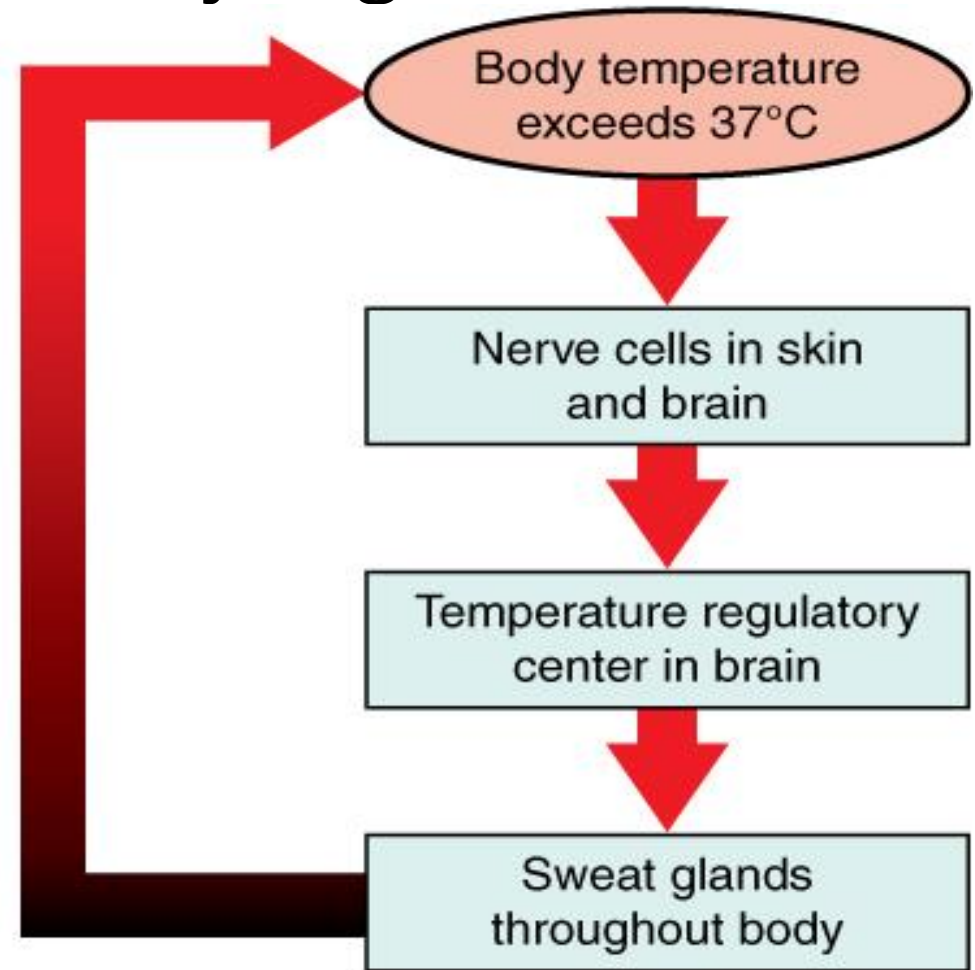
- o Examples include models of a bank, a processor



Body temperature is regulated by negative feedback



(a) Negative feedback loop



(b) Body temperature regulation



Temperature regulation feedback system works by promoting either heat loss or heat gain.

When the brain's temperature regulation center receives data from sensors indicating that body's temperature exceeds its normal range, it stimulates a cluster of brain cells referred to as the “heat-loss center.”



In contrast, activation of brain's heat-gain center by exposure to cold reduces blood flow to the skin, and blood returning from limbs is diverted into a network of deep veins. This arrangement traps heat closer to the body core and restricts heat loss.

<https://opentextbc.ca/anatomyandphysiology/chapter/1-5-homeostasis/>



Making a Washing Machine Closed Loop system

- Sound level detection techniques can be used to determine the point during water fill that the clothes have become totally immersed in the fluid.
- Determination of wash time and duration of agitation based on sensed wash fluid turbidity (optical sensing) and fill level.

<https://patents.google.com/patent/US5272892A/en>